

Mechanika Tuhých a Poddajných Telies

(MTPT)

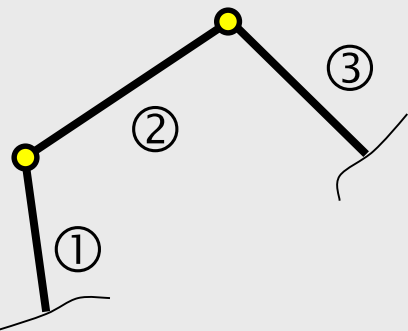
P-07

KINEMATIKA SÚSTAVY TELIES

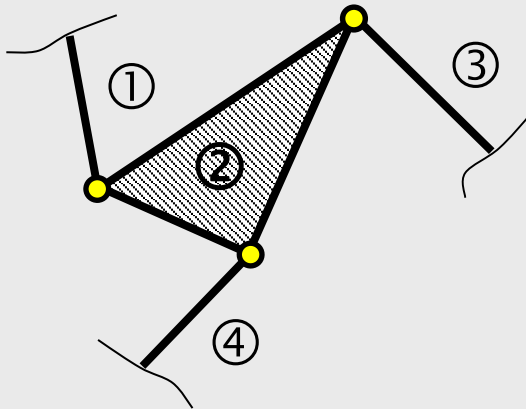
Množina telies (členov) vzájomne spojených väzbami (kinematickými dvojicami) do tzv. kinematických reťazí.

ČLENY (telesá)

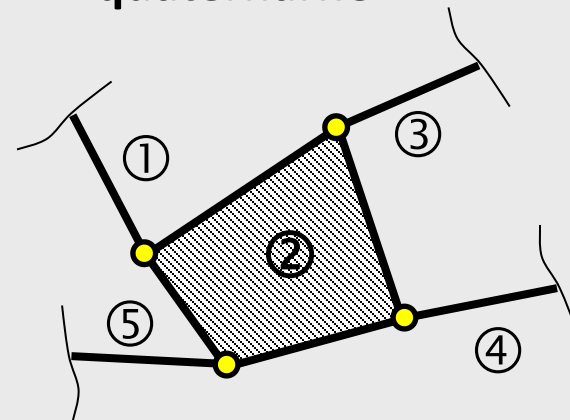
binárne



ternárne



quaternárne



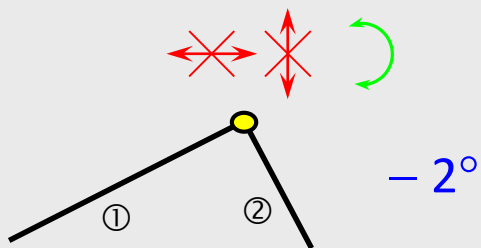
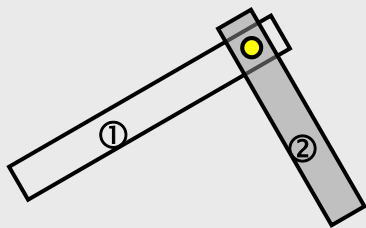
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KINEMATIKA SÚSTAVY TELIES

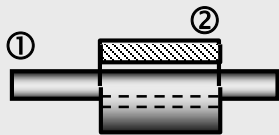
KINEMATICKÉ DVOJICE (väzby)

nižšie:

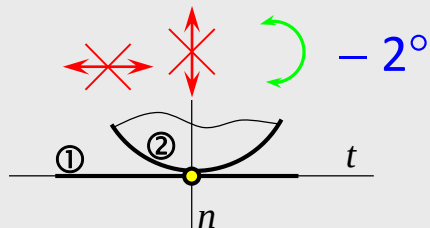
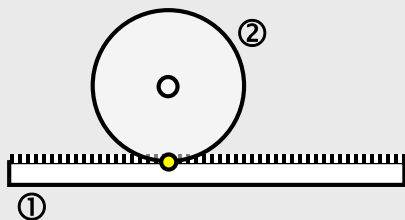
rotačná



posuvná

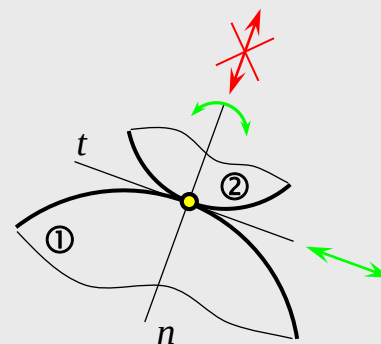


valivá



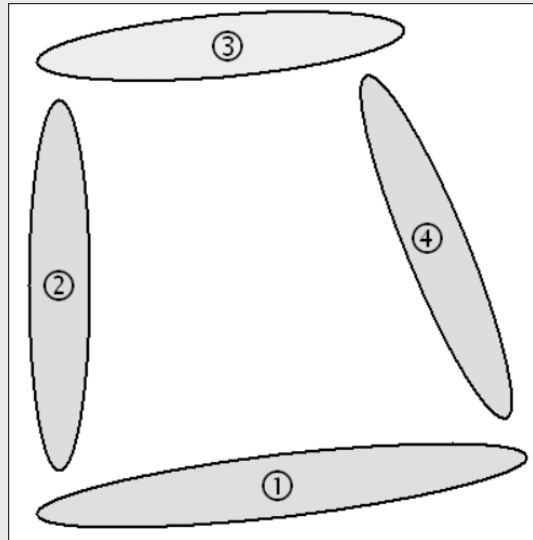
vyššie:

šmyková

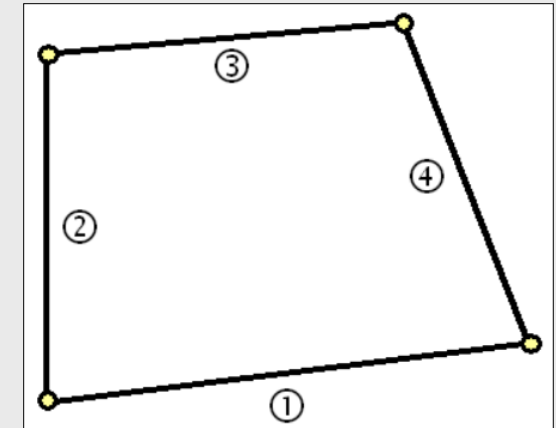
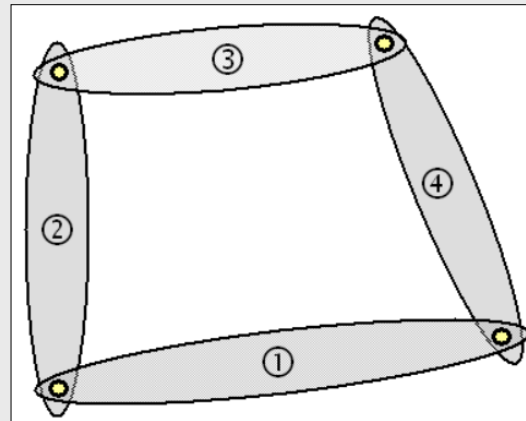


-1°

KINEMATIKA SÚSTAVY TELIES - pohyblivosť



$$n = 3 \cdot m = 3 \cdot 4 = 12$$



$$n = 3 \cdot m - \sum_{j=1}^2 j \cdot r_j = 3 \cdot m - (1 \cdot r_1 + 2 \cdot r_2) = 3 \cdot 4 - (1 \cdot 0 + 2 \cdot 4) = 4$$

$m = 4$ - počet telies v sústave

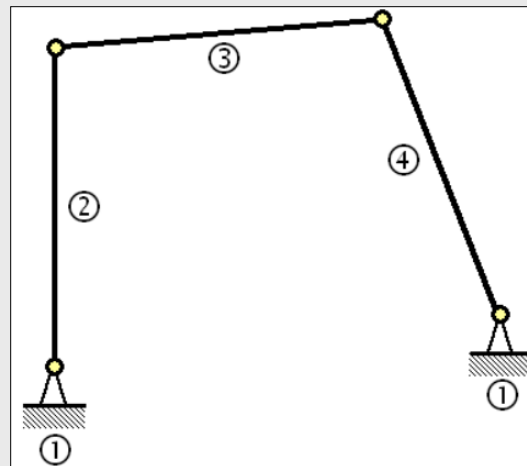
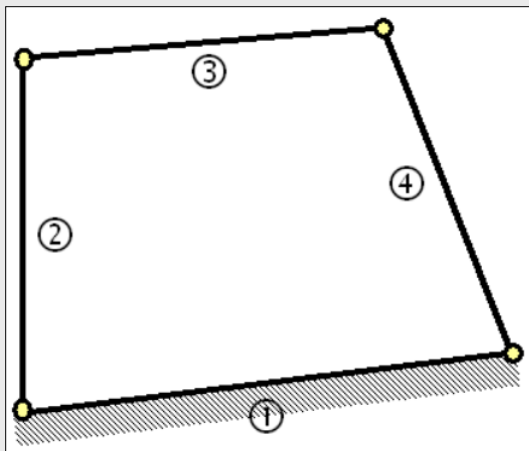
1° → $r_1 = 0$ väzieb (kinematické dvojice)

2° → $r_2 = 4$ väzby (kinematické dvojice)

$$n = 3 \cdot (m - 1) - \sum_{j=1}^2 j \cdot r_j = 3 \cdot (4 - 1) - (1 \cdot 0 + 2 \cdot 4) = 1$$

⇒ pohyblivá sústava telies

“mechanizmus”



$$n = 3 \cdot m - \sum_{j=1}^2 j \cdot r_j - 3 \Rightarrow n = 3 \cdot (m - 1) - \sum_{j=1}^2 j \cdot r_j$$

$$n = 3 \cdot (m - 1) - \sum_{j=1}^2 j \cdot r_j$$

počet väzieb odoberajúcich j° voľnosti

počet $^{\circ}$ voľnosti, ktoré odoberajú väzby v rovine

počet telies v sústave telies vrátane rámu

počet $^{\circ}$ voľnosti viazanej sústavy telies v rovine

Metódy riešenia mechanizmov

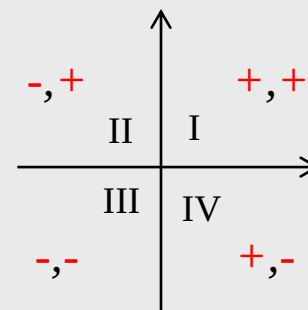
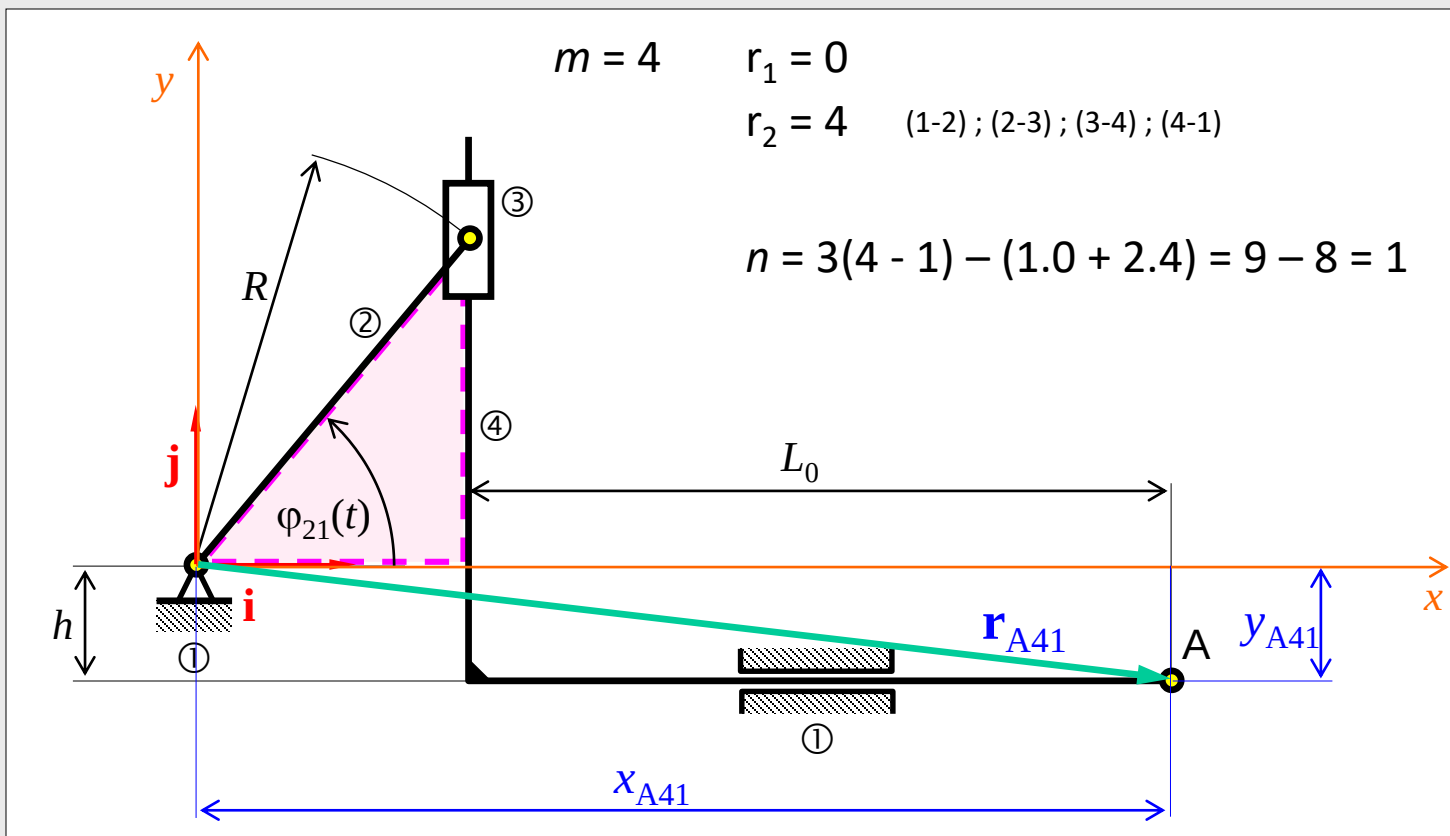
- trigonometrická metóda (goniometrické funkcie, trojuholník)
- vektorová metóda
- metóda komplexných čísel
- metóda prevodových funkcií
- maticové metódy

Trigonometrická metóda

Príklad:

Dané: $h, R, L_0, \varphi_{21}(t) = \pi t^2$

Určiť: $\mathbf{r}_{A41}$; $\mathbf{v}_{A41}$; $\mathbf{a}_{A41}$



$$x_{A41} = L_0 + R \cdot \cos(\varphi_{21})$$

$$y_{A41} = h$$

$$\mathbf{r}_{A41} = x_{A41} \mathbf{i} + y_{A41} \mathbf{j} = + [L_0 + R \cdot \cos(\pi t^2)] \mathbf{i} - [h] \mathbf{j}$$

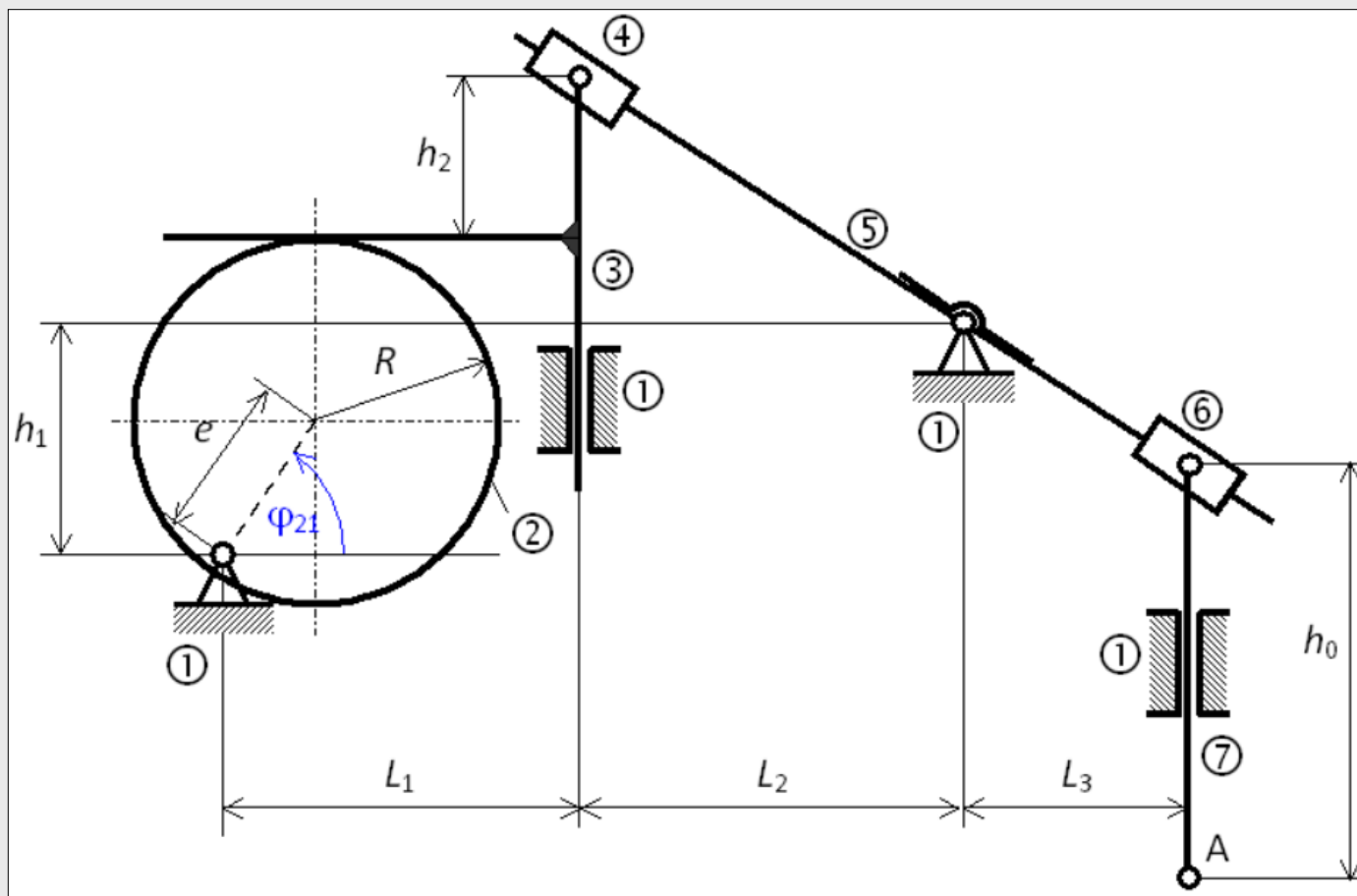
Trigonometrická metoda

$$\mathbf{r}_{A41} = [L_0 + R\cos(\pi t^2)]\mathbf{i} - [h]\mathbf{j}$$

$$\mathbf{v}_{A41} = \frac{d\mathbf{r}_{A41}}{dt} = -R(2\pi t)\sin(\pi t^2)\mathbf{i}$$

$$\mathbf{a}_{A41} = \frac{d\mathbf{v}_{A41}}{dt} = -2\pi R[\sin(\pi t^2) + 2\pi t^2 \cos(\pi t^2)]\mathbf{i}$$

KINEMATIKA

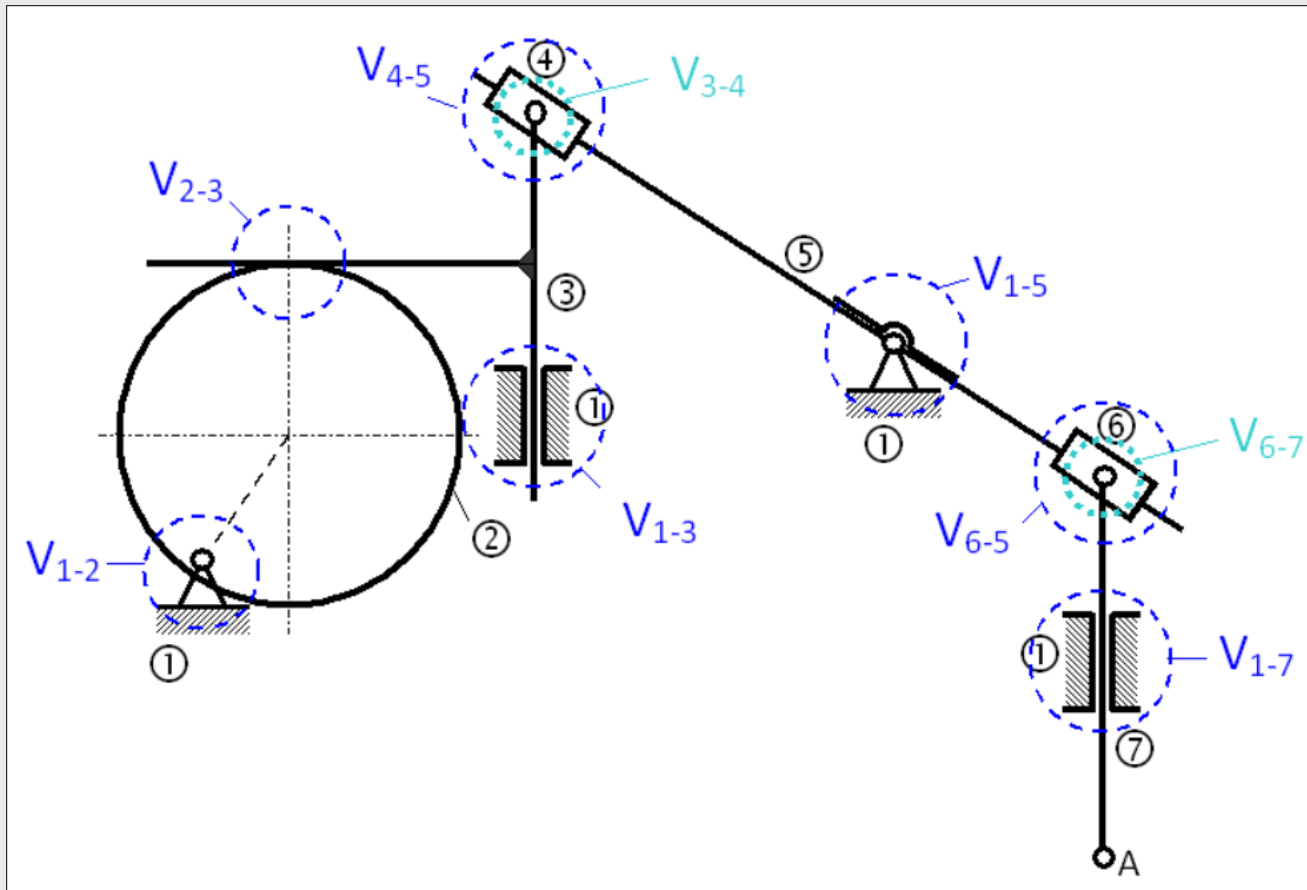


Vyjadrite vektor rýchlosti a vektor zrýchlenia bodu A.

Dané: $e, R, L_1, L_2, L_3, h_0, h_1, h_2, \varphi_{21} = \varphi_{21}(t)$

Určiť: $\mathbf{v}_{A71} = ?$ $\mathbf{a}_{A71} = ?$

Pohyblivosť sústavy telies



Označenie väzby	Typ väzby	Odobraté °voľnosti
V_{1-2}	RRK	2°
V_{1-5}	RRK	2°
V_{3-4}	RRK	2°
V_{6-7}	RRK	2°
V_{1-3}	PV	2°
V_{4-5}	PV	2°
V_{6-5}	PV	2°
V_{1-7}	PV	2°
V_{2-3}	ŠV	1°

RRK - rovinný rotačný kĺb

PV - posuvná väzba

ŠV - šmyková väzba

Výpočet počtu stupňov voľnosti sústavy:

$$n = 3 \cdot (m - 1) - \sum_{j=1}^3 j \cdot r_j = 3 \cdot (7 - 1) - (1 \cdot 1 + 2 \cdot 8 + 3 \cdot 0) = 18 - 17 = 1$$

Sústava je pohyblivá - má 1° voľnosti - je to mechanizmus.

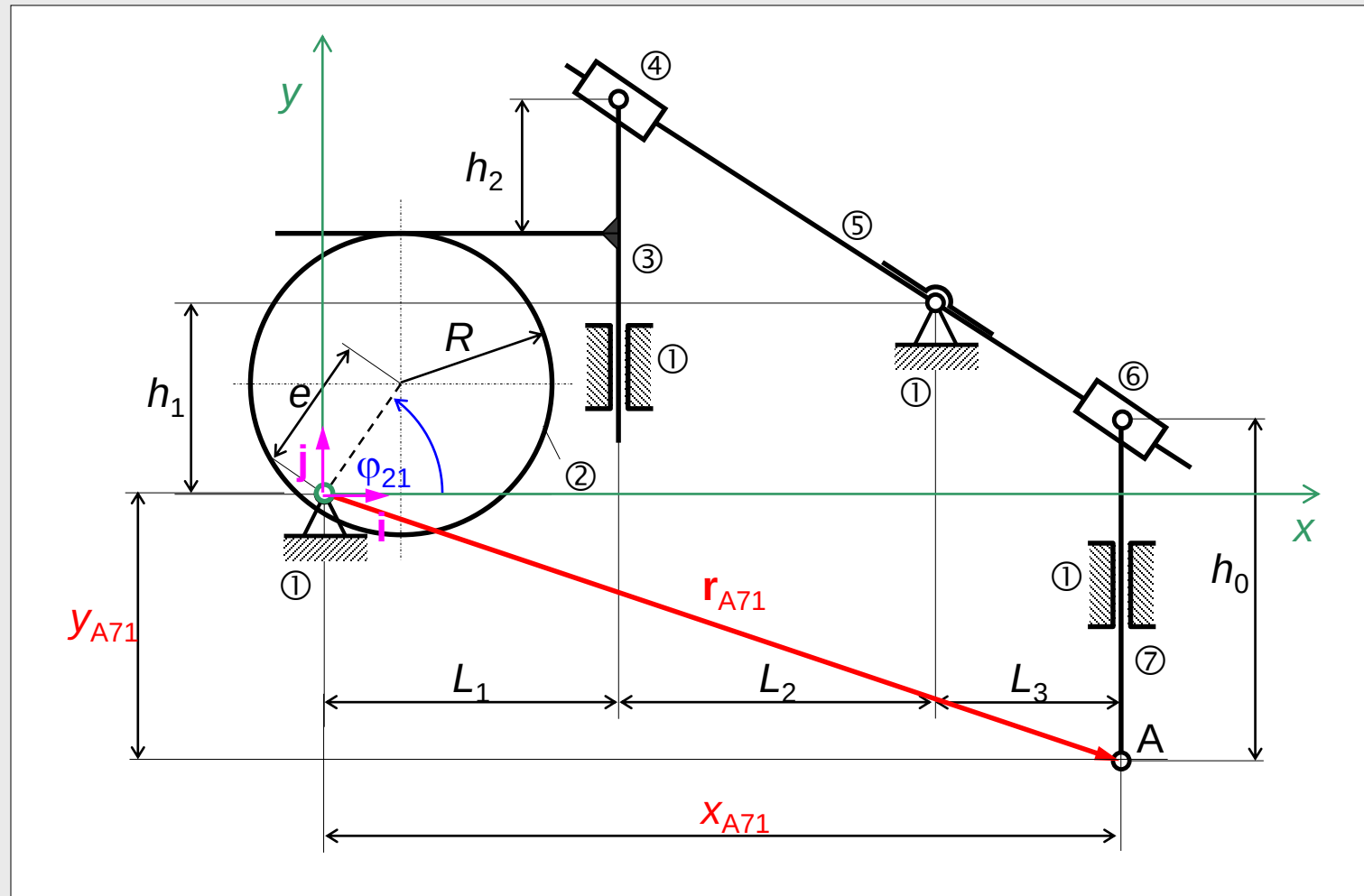
$$m = 7$$

$$j = 1 \quad r_1 = 1$$

$$j = 2 \quad r_2 = 8$$

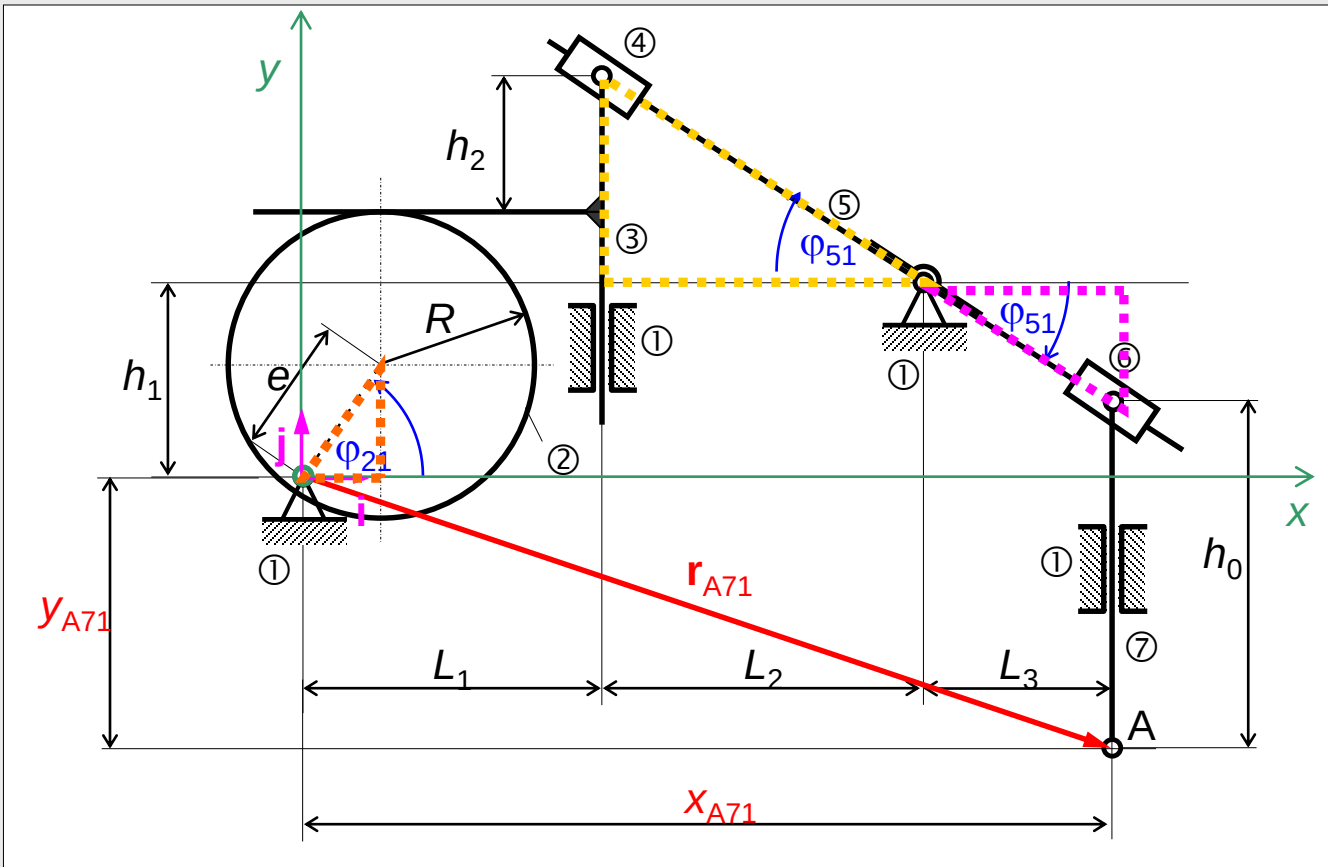
$$j = 3 \quad r_3 = 0$$

KINEMATIKA



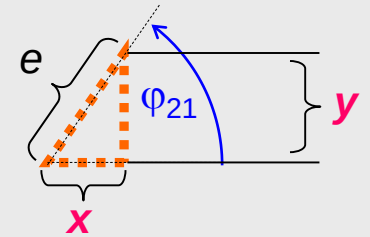
Metóda riešenia - trigonometrická metóda

KINEMATIKA

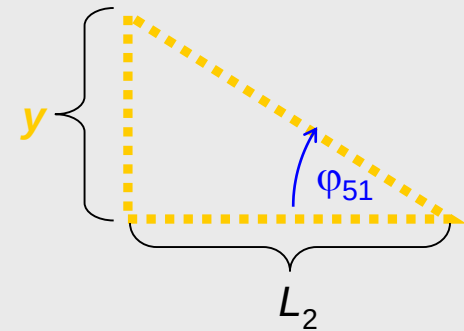


$$\tan \varphi_{51} = \frac{y}{L_2} = \frac{(h_2 + R + e \sin \varphi_{21}) - h_1}{L_2} \quad \varphi_{51} = \arctan \left[\frac{(h_2 + R + e \sin \varphi_{21}) - h_1}{L_2} \right]$$

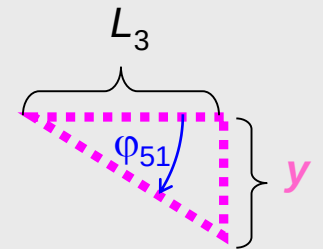
$$\mathbf{r}_{A71} = x_{A71}\mathbf{i} + y_{A71}\mathbf{j} \quad \begin{cases} x_{A71} = L_1 + L_2 + L_3 \\ y_{A71} = h_0 + y - h_1 = h_0 + L_3 \cdot \tan\varphi_{51} - h_1 \end{cases}$$



$$\begin{aligned}x &= e \cdot \cos \varphi_{21} \\y &= e \cdot \sin \varphi_{21}\end{aligned}$$

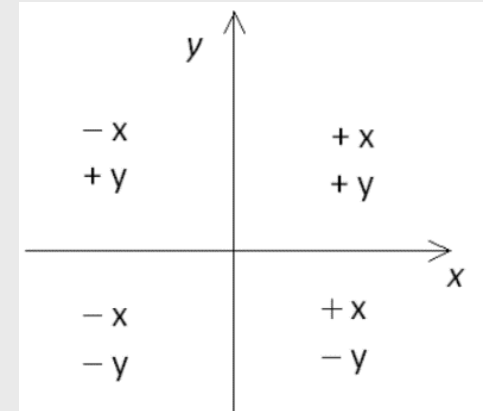
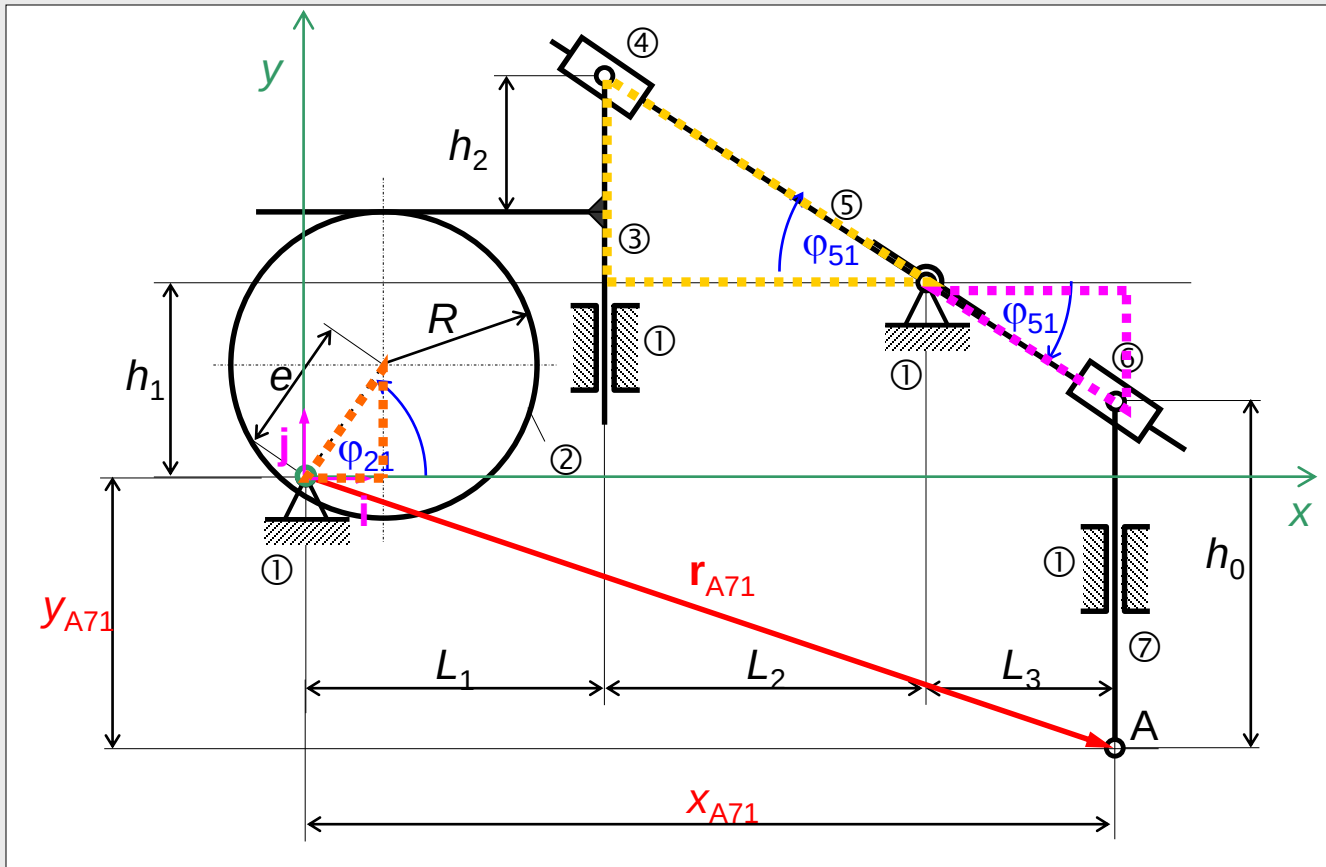


$$y = L_2 \cdot \tan \varphi_{51}$$



$$y = L_3 \cdot \tan \varphi_{51}$$

KINEMATIKA



$$\mathbf{r}_{A71} = +[L_1 + L_2 + L_3]\mathbf{i} - [h_0 + L_3 \tan \varphi_{51} - h_1]\mathbf{j}$$

$$\mathbf{r}_{A71} = [L_1 + L_2 + L_3]\mathbf{i} - [h_0 + L_3 \tan \varphi_{51} - h_1]\mathbf{j}$$

KINEMATIKA

$$\mathbf{r}_{A71} = [L_1 + L_2 + L_3]\mathbf{i} - [h_0 + L_3 \tan \varphi_{51} - h_1]\mathbf{j}$$

$$\mathbf{v}_{A71} = \frac{d\mathbf{r}_{A71}}{dt} = \dot{\mathbf{r}}_{A71} = \frac{d[L_1 + L_2 + L_3]}{dt}\mathbf{i} - \frac{d[h_0 + L_3 \tan \varphi_{51} - h_1]}{dt}\mathbf{j}$$

$$\mathbf{v}_{A71} = -L_3 \frac{1}{\cos^2 \varphi_{51}} \dot{\varphi}_{51} \mathbf{j}$$

$$\mathbf{a}_{A71} = \frac{d\mathbf{v}_{A71}}{dt} = \frac{d}{dt} \left[-L_3 \frac{\dot{\varphi}_{51}}{\cos^2 \varphi_{51}} \right] \mathbf{j} = -L_3 \frac{\ddot{\varphi}_{51} \cos^2 \varphi_{51} - \dot{\varphi}_{51} 2 \cos \varphi_{51} (-\sin \varphi_{51}) \dot{\varphi}_{51}}{\cos^4 \varphi_{51}} \mathbf{j}$$

$$\mathbf{a}_{A71} = -L_3 \frac{\ddot{\varphi}_{51} \cos \varphi_{51} + 2 \dot{\varphi}_{51}^2 \sin \varphi_{51}}{\cos^3 \varphi_{51}} \mathbf{j}$$

$$\dot{\varphi}_{51} = ?$$

$$\ddot{\varphi}_{51} = ?$$

KINEMATIKA

$$\varphi_{51} = \arctan \left[\frac{(h_2 + R + e \sin \varphi_{21}) - h_1}{L_2} \right]$$

$$\dot{\varphi}_{51} = \frac{1}{1 + \left[\frac{(h_2 + R + e \sin \varphi_{21}) - h_1}{L_2} \right]^2} \left(\frac{e}{L_2} \cos \varphi_{21} \right) \dot{\varphi}_{21}$$

$$\dot{\varphi}_{51} = \frac{\frac{e}{L_2} \cos \varphi_{21}}{\frac{L_2^2}{L_2^2 + [(h_2 + R + e \sin \varphi_{21}) - h_1]^2}} \dot{\varphi}_{21} = \frac{L_2 e \cos \varphi_{21}}{L_2^2 + [(h_2 + R + e \sin \varphi_{21}) - h_1]^2} \dot{\varphi}_{21}$$

$$\dot{\varphi}_{51} = \frac{L_2 e \cos \varphi_{21}}{L_2^2 + [(h_2 + R + e \sin \varphi_{21}) - h_1]^2} \dot{\varphi}_{21}$$

$$\ddot{\varphi}_{51} = \frac{d\dot{\varphi}_{51}}{dt} = \dots$$

$$\varphi_{21} = \varphi_{21}(t)$$

$$\dot{\varphi}_{21} = \frac{d\varphi_{21}}{dt} = \dots$$

$$\ddot{\varphi}_{21} = \frac{d\dot{\varphi}_{21}}{dt} = \dots$$